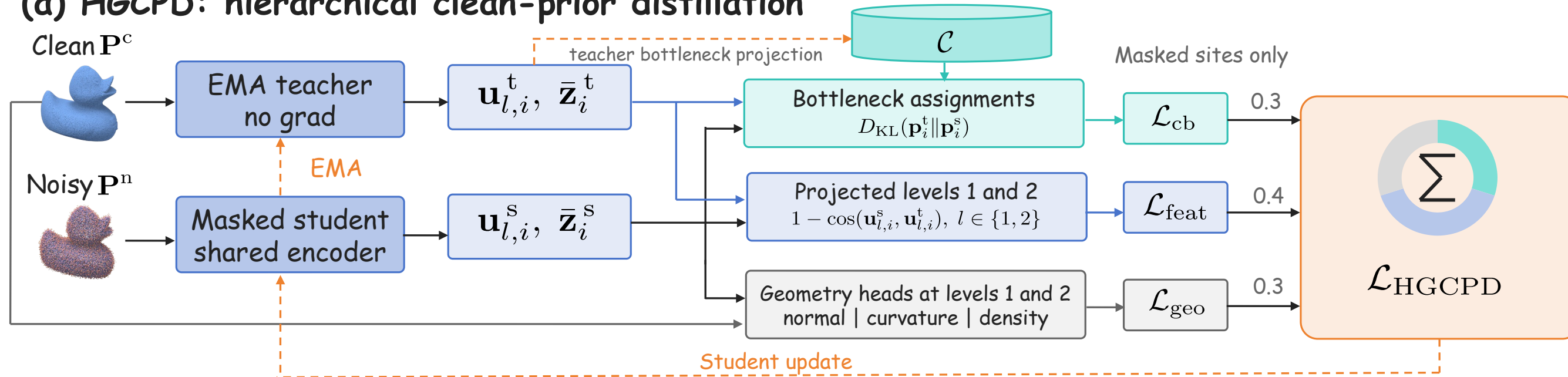
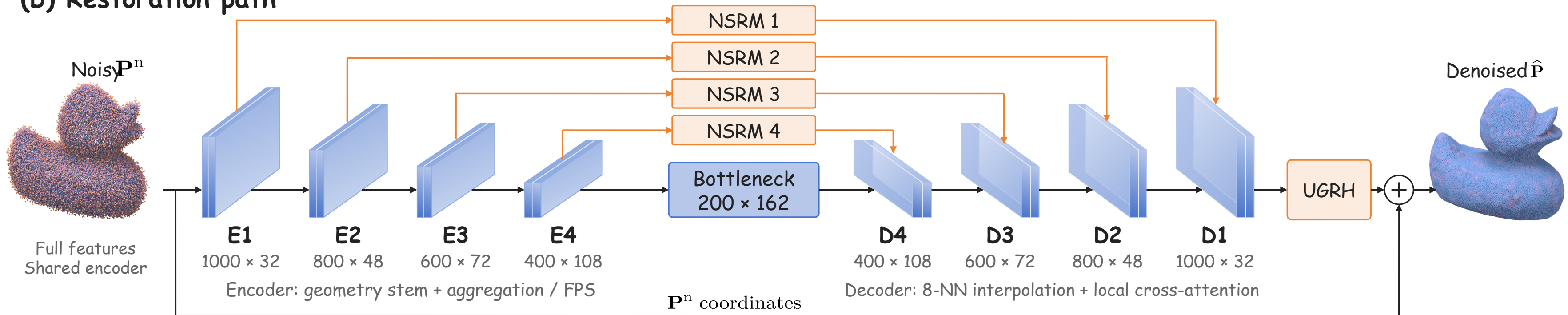


(a) HGCPD: hierarchical clean-prior distillation

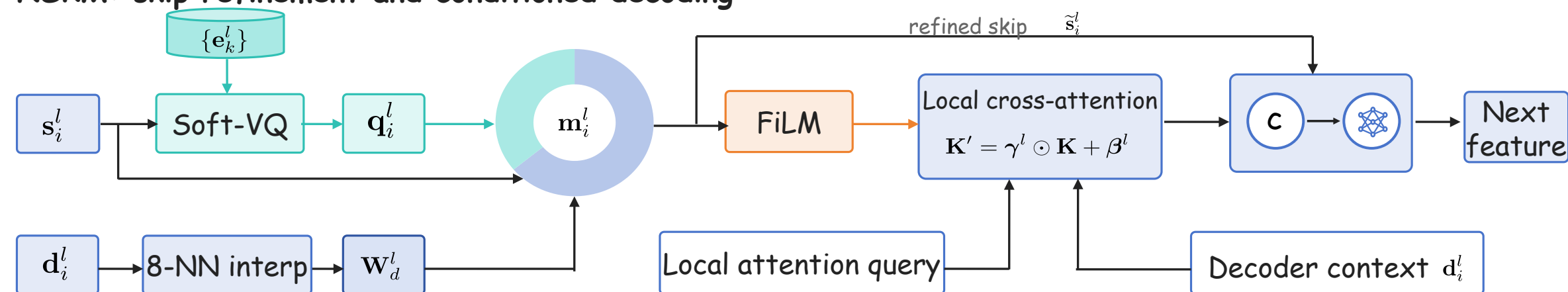


(b) Restoration path



(c) NASR: two-level adaptive restoration

NSRM: skip refinement and conditioned decoding



UGRH: coordinate control

